Q1

a) We have Complete Markets iff we can find a unique matrix Ω s.t. $X\Omega = I$

$X = \begin{bmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{bmatrix}$ $\Omega = \begin{bmatrix} 1 & 1 & 1 \\ w_1 & w_2 & w_3 \\ 1 & 1 & 1 \end{bmatrix}$, w_n is a portfolio weight vector using A, B and C.

$\begin{pmatrix} 10 \\ 20 \\ 30 \end{pmatrix}$ is asset A's payoff across the three states.

$X\Omega = I$

$Xw_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ $Xw_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ $Xw_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$Xw_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ is referring to can we combine A, B and C into a portfolio that pays \$1 if state 1 happens, but \$0 if state 2 or state 3 happens?

Because X is 3×3, $X\Omega = I$ has unique solution iff X is invertible

X invertible $\Leftrightarrow$ $\det(X) \neq 0$ $\Leftrightarrow$ $\text{rank}(X) = 3$

$X = \begin{bmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{bmatrix} = 10 \begin{bmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix}$

row 3 - row 2

$\begin{bmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

row 2 and row 1 exchange position

$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

row 2 - row 1

$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$

row 3 - row 1 - row 2

$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Thus, $\text{rank}(X) = 3$, X is invertible.

$\Omega = X^{-1}$ exists and is unique $X\Omega = XX^{-1} = I$

Thus, the market is complete

1b) Let X reflect the matrix of $N=3$ non-redundant asset payoffs.

Let P reflect the price vector for the original asset

Let P_c reflect the price vector for the contingent claims. $P_c = \begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix}$

use LOP to solve for P_c as a function of X and P.

$P = \begin{pmatrix} 16 \\ 8 \\ 6 \end{pmatrix}$ $X = \begin{bmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{bmatrix}$

LOP shows that two assets with identical payoffs in every state of nature must have the same price.

Therefore, $X^T P_c = P$

$$\begin{bmatrix} 10 & 20 & 30 \\ 10 & 10 & 10 \\ 0 & 10 & 10 \end{bmatrix}\begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = \begin{pmatrix} 16 \\ 8 \\ 6 \end{pmatrix}$$

Dividing everything by 10

$$\begin{cases} q_1 + 2q_2 + 3q_3 = 1.6 \\ q_1 + q_2 + q_3 = 0.8 \\ q_2 + q_3 = 0.6 \end{cases}$$

$$\Rightarrow \begin{cases} q_1 = 0.2 \\ q_2 = 0.4 \\ q_3 = 0.2 \end{cases}$$

Thus, $P_C = \begin{pmatrix} 0.2 \\ 0.4 \\ 0.2 \end{pmatrix}$

1c) $P_D = P_{Payoff(D)} \cdot P_C = \begin{bmatrix} 30 & 20 & 10 \end{bmatrix}\begin{pmatrix} 0.2 \\ 0.4 \\ 0.2 \end{pmatrix} = 6 + 8 + 2 = 16$

1d) $P(X) = E[MX]$

$$P(X) = \sum_{s=1}^{S} \pi(s) M(s) X(s)$$

$q_s = \pi(s) M(s)$

$$M(s) = \frac{q_s}{\pi(s)}$$

$P_C = \begin{pmatrix} 0.2 \\ 0.4 \\ 0.2 \end{pmatrix}$ $\quad \pi(s) = \begin{pmatrix} \frac{1}{4} \\ \frac{1}{2} \\ \frac{1}{4} \end{pmatrix}$

$M(s) = \begin{pmatrix} 0.8 \\ 0.8 \\ 0.8 \end{pmatrix}$

Since by definition, no arbitrage means that

① An asset with Payoff ≥ 0, must have Price ≥ 0

② An asset with Payoff ≥ 0 and a chance for Payoff > 0 must have Price > 0

We have $X \geq 0$, $P_r(X > 0) > 0$, $P(X) = E[MX] > 0$

Thus, there is no arbitrage.

$P(X) = E[MX] = 0.8\,E[X]$

The SDF across three states stays constant, this implies it is risk-neutral

1e) $P_D = E[MX_D] = \sum_{s=1}^{3} \pi(s) M(s) X_D(s)$

$= \frac{1}{4} \cdot 0.8 \cdot 30 + \frac{1}{2} \cdot 0.8 \cdot 20 + \frac{1}{4} \cdot 0.8 \cdot 10$

$= 6 + 8 + 2 = 16$

This is the same as in 1c.